



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY TIRUCHIRAPPALLI
(AN INSTITUTE OF NATIONAL IMPORTANCE)
Sethurapatti, Tiruchirappalli - 620012
Department of Computer Science and Engineering
AICTEQIP-PG Program on "Machine Learning and Cyber physical systems"

Lab 1: Introduction to Data and Basic Data Handling

Objective: Learn to load, inspect, and manipulate structured healthcare data using Pandas.

Dataset: Pima Indians Diabetes

Exercise 1.1: Load the dataset and print the shape (number of rows and columns).

Exercise 1.2: Display the column names and their data types.

Exercise 1.3: Print the first 5 rows of the dataset.

Exercise 1.4: Count the number of unique values in each column.

Exercise 1.5: Filter the dataset to include only rows where:

- `glucose > 120`
- `bmi > 30`

Exercise 1.6: Create a new column `bmi_age_ratio` by dividing BMI by age.

Exercise 1.7: Calculate the average `bmi_age_ratio` for each class label (`outcome`).

Exercise 1.8: Display the count of each class label in the `outcome` column.

Exercise 1.9: Sort the dataset in descending order based on `glucose`.

Lab 2: Data Quality Assessment and Data Acquisition



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY TIRUCHIRAPPALLI
(AN INSTITUTE OF NATIONAL IMPORTANCE)
Sethurapatti, Tiruchirappalli - 620012
Department of Computer Science and Engineering
AICTEQIP-PG Program on "Machine Learning and Cyber physical systems"

Objective: Assess data quality, handle zero-value entries, identify duplicates, and fetch additional financial data.

Exercise 2.1: Load dataset (Pima Diabetes dataset)

Exercise 2.2: Check and report the number of duplicate rows.

Exercise 2.3: Count the number of zero entries in the following columns (treat them as missing):

- glucose, bloodpressure, skinthickness, insulin, bmi

Exercise 2.4: Use the IQR method to detect outliers in the **bmi** column.

Exercise 2.5: Remove the identified outliers and report the new shape of the dataset.

Exercise 2.6: Use the **yfinance** library to download daily S&P 500 index data for the last 1 year.

Exercise 2.7: Filter and save only those rows where the trading volume is higher than the average volume.



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY TIRUCHIRAPPALLI
(AN INSTITUTE OF NATIONAL IMPORTANCE)
Sethurapatti, Tiruchirappalli - 620012
Department of Computer Science and Engineering
AICTEQIP-PG Program on "Machine Learning and Cyber physical systems"

Lab 3: Handling Missing Values

Objective: Detect and impute missing values using manual and automated techniques.

Exercise 3.1: Replace zeros with NaN for the relevant columns

Exercise 3.2: Impute the missing values using the mean (manually).

Exercise 3.3: Impute the missing values using `SimpleImputer` with the `median` strategy.

Exercise 3.4: Create a function `clean_pima()` that:

- Standardizes column names
- Removes duplicates
- Imputes missing values
- Removes outliers in the `age` and `bmi` columns

Exercise 3.5: Save the cleaned dataset to a CSV file `cleaned_pima.csv`

Note: These exercises use standard Python libraries including `pandas`, `numpy`, `sklearn`, and `yfinance`. Ensure they are installed in your environment.



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY TIRUCHIRAPPALLI

(AN INSTITUTE OF NATIONAL IMPORTANCE)

Sethurapatti, Tiruchirappalli - 620012

Department of Computer Science and Engineering

AICTEQIP-PG Program on "Machine Learning and Cyber physical systems"

Lab 4: Introduction to Statistical analysis tool - JASP

Dataset: <https://archive.ics.uci.edu/dataset/320/student+performance>

Objectives:

By the end of this lab, participants will be able to:

- Explore and summarize educational data using descriptive statistics
- Perform group comparisons using t-tests and ANOVA
- Investigate relationships using correlation and linear regression
- Interpret statistical results and communicate findings clearly

Lab Experiments

4.1. Experiment 1: Descriptive Statistics and Group Comparisons

Objective: Explore central tendencies, data spread, and gender-based differences in final scores.

Tasks:

1. Load the student dataset into JASP.
2. Use **Descriptives** to summarize **G3** (final grade) for each gender.
3. Use **T-Tests** → **Independent Samples T-Test**:
 - Grouping Variable: **gender**
 - Test Variable: **G3**
 - Check assumptions (Levene's test)
4. Record and interpret:
 - Means by group



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY TIRUCHIRAPPALLI
(AN INSTITUTE OF NATIONAL IMPORTANCE)
Sethurapatti, Tiruchirappalli - 620012
Department of Computer Science and Engineering
AICTEQIP-PG Program on "Machine Learning and Cyber physical systems"

- p-value
 - Whether gender has a significant impact on G3
-

4.2. Experiment 2: Effect of Study Time on Academic Performance

Objective: Test whether study time categories significantly affect final exam scores.

Tasks:

1. Use ANOVA → One-Way ANOVA:
 - Factor: study_time
 - Dependent Variable: G3
 - Enable Descriptives and Homogeneity test
 2. Interpret:
 - p-value
 - Whether assumptions of homogeneity are satisfied
 3. Determine if post hoc analysis is needed (optional).
-

4.3. Experiment 3: Correlation and Regression Analysis

Objective: Examine relationships among grade variables and predict performance using regression.

Tasks:



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY TIRUCHIRAPPALLI

(AN INSTITUTE OF NATIONAL IMPORTANCE)

Sethurapatti, Tiruchirappalli - 620012

Department of Computer Science and Engineering

AICTEQIP-PG Program on "Machine Learning and Cyber physical systems"

1. Use **Regression** → **Correlation Matrix**:

- Variables: **G1, G2, G3, absences**
- Enable significance tests and scatter plots

2. Use **Regression** → **Linear Regression**:

- Dependent Variable: **G3**
- Covariate: **G2**

3. Record and interpret:

- Correlation coefficients (r) and their significance
- R-squared and regression coefficients (β)
- Regression equation: **$G3 = \beta_0 + \beta_1 \times G2$**
- Predicted G3 value for G2 = 15